

Julianne Zhu

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EDUCATION

University of Waterloo

Waterloo, ON

BCS, Honours Computer Science with Co-operative Program, GPA: 3.7/4.0

Sep. 2025 – Apr. 2030

- President Scholarship of Distinction
- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Functional Programming, Logic & Computation, Software Development Tools & Techniques

EXPERIENCE

Network Engineering Intern

May 2026 – Sep. 2026

ORBCOMM, Network Services Department

Ottawa, ON

- Developed 8 Grafana dashboards with **MySQL** data pipelines to monitor satellite network performance across 12 metrics, cutting load times by **86%** through dynamic filtering and optimized SQL queries
- Shipped an AI-driven analytics service to extend predictive forecasting by 60x, cut pipeline runtime by 78%, and lower false-positive anomaly alerts by 80% using **FastAPI**, APScheduler, PyMySQL, Jinja2, and **MariaDB**
- Launched the service internally for 7 engineering teams by presenting to leadership with a system design document and live demo, pushing production code to the company **GitHub** repository

Undergraduate Research Mentee

Jan. 2026 – Present

Usable Security & Privacy Research, University of Waterloo

Waterloo, ON

- Reached **500+** attendees from 30+ institutions at CAN-CWiC and IEEE S&P by co-authoring a research paper on how users seek security and privacy advice from Large Language Models (LLMs)
- Evaluated 2 LLMs against 100 security and privacy prompts across 15 categories by building a validated **54-database** golden-answer dataset and scoring rubric with 3 security and HCI researchers

PROJECTS

CaneOS | Swift, Python, C++, SwiftUI, MongoDB Atlas, Auth0, Gemini API, Vercel, Websockets

Jul. 2026

Full Stack & Embedded Developer, Hack the 6ix

Toronto, ON

- Built a clip-on white cane module for visually impaired users, pairing ToF sensors with 30 FPS YOLOv6 camera detection on an **Arduino UNO Q** within a 4m hazard-detection range, feeding a **Python**/FastAPI pipeline
- Classified hazards via Gemini vision and narrated results within 3 seconds via ElevenLabs, reaching **98% accuracy** through the pipeline's real-time on-device inference, giving users time to react safely
- Developed companion **SwiftUI** iPhone and Apple Watch apps with directional haptics, 4-tier SOS alerts, and Auth0-secured sync of incidents and contacts to **MongoDB Atlas** via a Node.js serverless proxy

TrulyHer | React, TypeScript, PostgreSQL, pgvector, Node.js, PyTorch, ONNX, Vercel

Sep. 2025 – Aug. 2026

AI Integration & Full Stack Developer, TechNova

Waterloo, ON

- Built a full-stack **Retrieval-Augmented Generation (RAG)** application to help women in CS manage imposter syndrome, enabling semantic search across **200+ journal entries** via HNSW vector embeddings and generating personalized reports across 5 analytics views using **React**, Node.js, and **PostgreSQL** with **pgvector**
- Detected imposter syndrome at **0.92 F1** on 200 hand-labeled entries using an ML pipeline that trains a **DistilBERT** classifier, calibrates its confidence scores, and exports to **ONNX** with INT8 quantization

Project Tech Careers | JavaScript, HTML, CSS, VS Code

May – Jun. 2024

Platform Designer and UI/UX Developer, PTCHacks

Ottawa, ON

- Designed a 4-stage platform, curating **100+ resources**, to help women in CS find mentorship, scholarships, programs, and community resources across different stages of their education using **JavaScript** and HTML
- Awarded the **Gender Equality Track Award** over 45 competing teams for innovation, design, and impact in addressing the United Nations Sustainable Development Goals

TECHNICAL SKILLS

Languages: Python, C/C++, Java, JavaScript, TypeScript, HTML/CSS, SQL, Bash, Swift, R

Frameworks and APIs: React, Node.js, REST APIs, PyTorch, SQLAlchemy, Gemini API, Uvicorn, Resend API

Data and Analytics: PostgreSQL, MongoDB Atlas, pgvector, InfluxDB, pandas, NumPy, Grafana, scikit-learn

Tools and Systems: Git/GitHub, Linux, AWS, Google Cloud Platform, Vercel, VS Code, Xcode, VirtualBox